



INT 422
Pattern Recognition

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Lecture 4-5: Agenda

- Bayes Classifiers Issues
- The Gaussian Density
- Discriminant Functions
- Why Go Fuzzy?
- The Fuzzy Way of Thinking
- The Concept of Fuzzy Logic
- Fuzzy logic applications
- Fuzzy System basic components
- Fuzzy inference system

Bayesian Decision Theory

- $p(\mathbf{x}\,|w_j)$ is the class-conditional probability density function.
- $P(w_j)$ is the prior probability that nature is in state w_j.
- The posterior probability can be computed as

$$P(w_j\,|\mathbf{x}) = \frac{p(\mathbf{x}\,|w_j)P(w_j)}{p(\mathbf{x})}$$

where $p(\mathbf{x}) = \sum_{j=1}^{c} p(\mathbf{x}\,|w_j)P(w_j)$.

Bayes Classifiers Issues

The zero-frequency problem

- One of the disadvantage with Naive-Bayes is that if you have **no occurrences** of a class label and a certain attribute value together then the frequency-based probability estimate will be zero.

	Outlook	Temperature	Humidity	Windy	Play Golf
0	Rainy	Hot	High	False	No
1	Rainy	Hot	High	True	No
2	Overcast	Hot	High	False	Yes
3	Sunny	Mild	High	False	Yes
4	Sunny	Cool	Normal	False	Yes
5	Sunny	Cool	Normal	True	No
6	Overcast	Cool	Normal	True	Yes
7	Rainy	Mild	High	False	No
8	Rainy	Cool	Normal	False	Yes
9	Sunny	Mild	Normal	False	Yes
10	Rainy	Mild	Normal	True	Yes
11	Overcast	Mild	High	True	Yes
12	Overcast	Hot	Normal	False	Yes
13	Sunny	Mild	High	True	No

Bayes Classifiers Issues

- **The zero-frequency problem:**
- **Solution** (Laplace estimator)

generalized Laplace estimate:

$$P(A_i = v_j | c_k) = \frac{n_{ijk} + 1}{n_k + s_i}$$

- n_{ijk}: number of examples in c_k where $A_i = v_j$
- n_k: number of examples in c_k
- s_i: number of possible values for A_i

$P(\text{Overcast} \mid No) = \frac{0}{5} = 0$ $\xrightarrow{\textit{Laplace estimator}}$ $P(\text{Overcast} \mid No) = \frac{0+1}{5+3} = \frac{1}{8}$

Bayes Classifiers Issues

- **Numerical Predictors**
  - Numerical variables need to be transformed to their categorical counterparts (binning) before constructing their frequency tables. The other option we have is using the distribution of the numerical variable to have a good guess of the frequency.
  - For example, one common practice is to assume normal distributions for numerical variables.
  - The probability density function for the normal distribution is defined by two parameters (**mean and standard deviation**).

Gaussian distribution

$$P(x_i|y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} exp\left(-\frac{(x_i-\mu_y)^2}{2\sigma_y^2}\right)$$

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$ Mean

$$\sigma = \left[\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\mu)^2\right]^{0.5}$$ Standard deviation

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$ Normal distribution

Bayes Classifiers Issues

- Example #1

		Humidity									Mean	StDev
Play Golf	yes	86	96	80	65	70	80	70	90	75	79.1	10.2
	no	85	90	70	95	91					86.2	9.7

$$P(\text{humidity} = 74 \mid \text{play} = \text{yes}) = \frac{1}{\sqrt{2\pi}(10.2)} e^{-\frac{(74-79.1)^2}{2(10.2)^2}} = 0.0344$$

$$P(\text{humidity} = 74 \mid \text{play} = \text{no}) = \frac{1}{\sqrt{2\pi}(9.7)} e^{-\frac{(74-86.2)^2}{2(9.7)^2}} = 0.0187$$

Example #2: Gender classification

- Problem: classify whether a given person is a male or a female based on the measured features. The features include height, weight, and foot size.

Person	height (feet)	weight (lbs)	foot size(inches)
male	6	180	12
male	5.92 (5'11")	190	11
male	5.58 (5'7")	170	12
male	5.92 (5'11")	165	10
female	5	100	6
female	5.5 (5'6")	150	8
female	5.42 (5'5")	130	7
female	5.75 (5'9")	150	9

$= \sigma^2$

Person	mean (height)	variance (height)	mean (weight)	variance (weight)	mean (foot size)	variance (foot size)
male	5.855	$3.5033 * 10^{-2}$	176.25	$1.2292 * 10^{2}$	11.25	$9.1667 * 10^{-1}$
female	5.4175	$9.7225 * 10^{-2}$	132.5	$5.5833 * 10^{2}$	7.5	1.6667

Testing

Below is a sample to be classified as male or female.

Person	height (feet)	weight (lbs)	foot size(inches)
sample	6	130	8

$$\text{posterior (male)} = \frac{P(\text{male})\, p(\text{height} \mid \text{male})\, p(\text{weight} \mid \text{male})\, p(\text{foot size} \mid \text{male})}{\textit{evidence}}$$

$$\text{posterior (female)} = \frac{P(\text{female})\, p(\text{height} \mid \text{female})\, p(\text{weight} \mid \text{female})\, p(\text{foot size} \mid \text{female})}{\textit{evidence}}$$

$P(\text{male}) = 0.5$

$$p(\text{height} \mid \text{male}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(6-\mu)^2}{2\sigma^2}\right) \approx 1.5789,$$

$$p(\text{weight} \mid \text{male}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(130-\mu)^2}{2\sigma^2}\right) = 5.9881 \cdot 10^{-6}$$

$$p(\text{foot size} \mid \text{male}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(8-\mu)^2}{2\sigma^2}\right) = 1.3112 \cdot 10^{-3}$$

$\text{posterior numerator (male)} = \text{their product} = 6.1984 \cdot 10^{-9}$

$P(\text{female}) = 0.5$

$p(\text{height} \mid \text{female}) = 2.2346 \cdot 10^{-1}$

$p(\text{weight} \mid \text{female}) = 1.6789 \cdot 10^{-2}$

$p(\text{foot size} \mid \text{female}) = 2.8669 \cdot 10^{-1}$

$\text{posterior numerator (female)} = \text{their product} = 5.3778 \cdot 10^{-4}$

Since posterior numerator is greater in the female case, we predict the sample is female.

The Gaussian Density

- Gaussian can be considered as a model where the feature vectors for a given class are continuous-valued, randomly corrupted versions of a single typical or prototype vector.

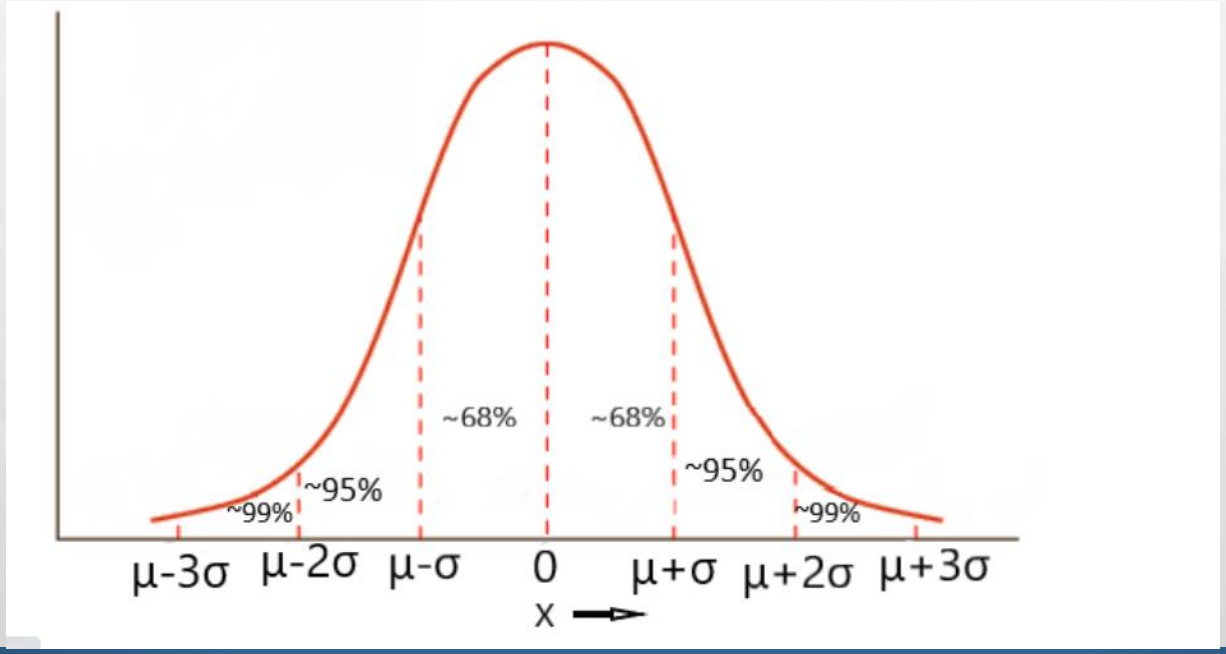


Figure 1: Gaussian distribution follows a bell-shaped curve and hence generally known as a bell curve.

The Gaussian Density

- Some properties of the Gaussian:
  - Analytically tractable.
  - Completely specified by the 1st and 2nd moments.
  - Has the maximum entropy of all distributions with a given mean and variance.
  - Many processes are asymptotically Gaussian (Central Limit Theorem).
  - Linear transformations of a Gaussian are also Gaussian.
  - Uncorrelatedness implies independence.

Discriminant Functions

- A useful way of representing classifiers is through *discriminant functions* $g_i(\mathbf{x}), i = 1, \ldots, c$, where the classifier assigns a feature vector $\mathbf{x}$ to class w_i if

$$g_i(\mathbf{x}) > g_j(\mathbf{x}) \quad \forall j \neq i.$$

- For the classifier that minimizes conditional risk

$$g_i(\mathbf{x}) = -R(\alpha_i|\mathbf{x}).$$

- For the classifier that minimizes error

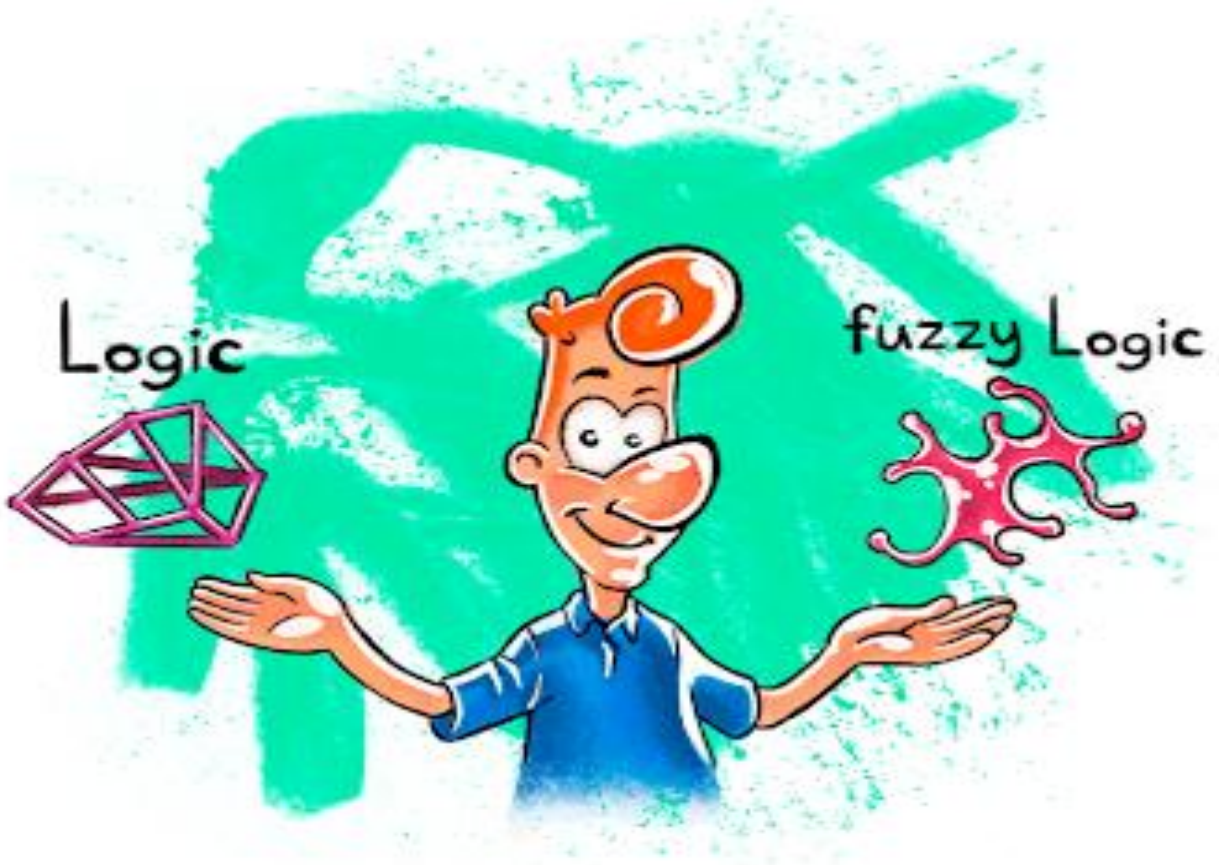
$$g_i(\mathbf{x}) = P(w_i|\mathbf{x}).$$

Discriminant Functions

- These functions divide the feature space into c *decision regions* ($R_1, \ldots, R_c$), separated by *decision boundaries*.
- Note that the results do not change even if we replace every $g_i(\mathbf{x})$ by $f(g_i(\mathbf{x}))$ where $f(\cdot)$ is a monotonically increasing function (e.g., logarithm).
- This may lead to significant analytical and computational simplifications.

Lecture 5

FUZZY LOGIC



Why Go Fuzzy?

- Fuzzy logic models human expertise and knowledge in some task or application
- Consider conventional binary logic
  - Variables may take values of TRUE or FALSE (0 or 1)
- Try then to answer a simple question with binary logic
  - What do you consider warm temperature?
- How to answer?
  - You could try to give a value or interval of “warm” temperature
  - But then when does the temperature become cold or hot?

Application Areas of Fuzzy Logic

Automotive Systems

1. Automatic Gearboxes
2. Four-Wheel Steering
3. Vehicle environment control

Application Areas of Fuzzy Logic

Consumer Electronic Goods

1. Hi-Fi Systems
2. Photocopiers
3. Still and Video Cameras
4. Television
5. Domestic Goods

Application Areas of Fuzzy Logic

- Domestic Goods

1. Microwave Ovens
2. Refrigerators
3. Toasters
4. Vacuum Cleaners
5. Washing Machines

Environment Control

1. Air Conditioners/Dryers/Heaters
2. Humidifiers

What is Fuzzy Logic?

- Fuzzy Logic (FL) is a method of reasoning that resembles human reasoning. The approach of FL imitates the way of decision making in humans that involves all intermediate possibilities between digital values YES and NO.
- The conventional logic block that a computer can understand takes precise input and produces a definite output as TRUE or FALSE, which is equivalent to human's YES or NO.
- The inventor of fuzzy logic, Lotfi Zadeh, observed that unlike computers, the human decision making includes a range of possibilities between YES and NO, such as −
- It can be implemented in hardware, software, or a combination of both.

Conventional set vs. fuzzy set

X: universe of discourse. Set A is defined by

$$A=\{ x \mid m(x)=T \wedge x \text{ in } X\}$$

where m(x) is a **membership function**.

- For conventional set, the range of m(x) is {T, F}
- For fuzzy set, the range of m(x) is [0,1]. So, the above definition cannot be used for fuzzy set. We cannot say clearly if x is in A or not

Examples of fuzzy set

- ✓ young people, old people, kind people
- ✓ temperature is hot, just good, a little bit cold

Description of fuzzy set

Membership function of a fuzzy set A: m_A: X->[0,1]

If the universe of discourse $X=\{x_1,x_2,\ldots,x_N\}$, A is described as follows:

$$A = m_A(x_1)/x_1 + m_A(x_2)/x_2 + \ldots + m_A(x_N)/x_N$$
$$= \mathrm{S} m_A(x_i)/x_i$$



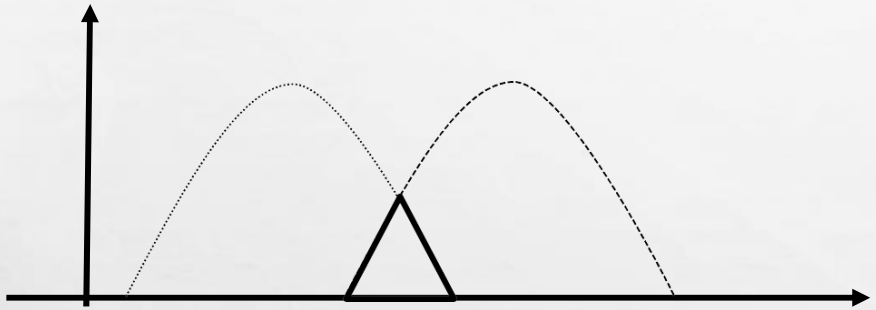
where / is a separator, and + is the logic OR.

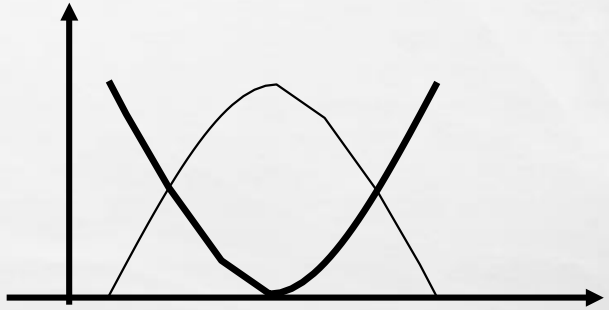
If X is a continuous space, integral is used instead of the summation.

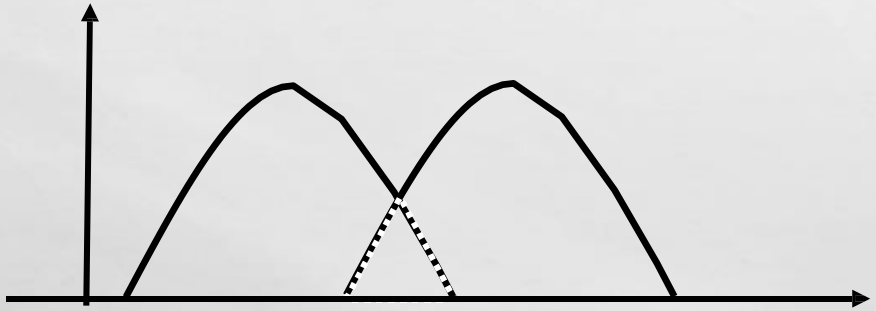
Operations of fuzzy sets

Meaning	Set notation	Logic notation	Membership function
Equivalence	$A = B$	$\mu_A(x) \Leftrightarrow \mu_B(x)$	$\mu_A(x) = \mu_B(x)$
Implication	$A \subseteq B$	$\mu_A(x) \Rightarrow \mu_B(x)$	$\mu_A(x) \leq \mu_B(x)$
Complement (negation)	$\overline{A}$	$\neg\mu_A(x)$	$\mu_{\overline{A}}(x) = 1 - \mu_A(x)$
Union, OR (disjunction)	$A \cup B$	$\mu_A(x) \vee \mu_B(x)$	$\mu_{A \cup B}(x) = \max\{\mu_A(x), \mu_B(x)\}$
Intersection, AND (conjunction)	$A \cap B$	$\mu_A(x) \wedge \mu_B(x)$	$\mu_{A \cap B}(x) = \min\{\mu_A(x), \mu_B(x)\}$

EXAMPLES







Upper-left: AND
Lower-left: OR
Upper-right: Negation

Example

The universe of discourse:

X={Masahiro, Tsuyoshi, Takuya, Saburo, Masami}

Fuzzy sets A="young persons" and B="Tall persons" are defined by

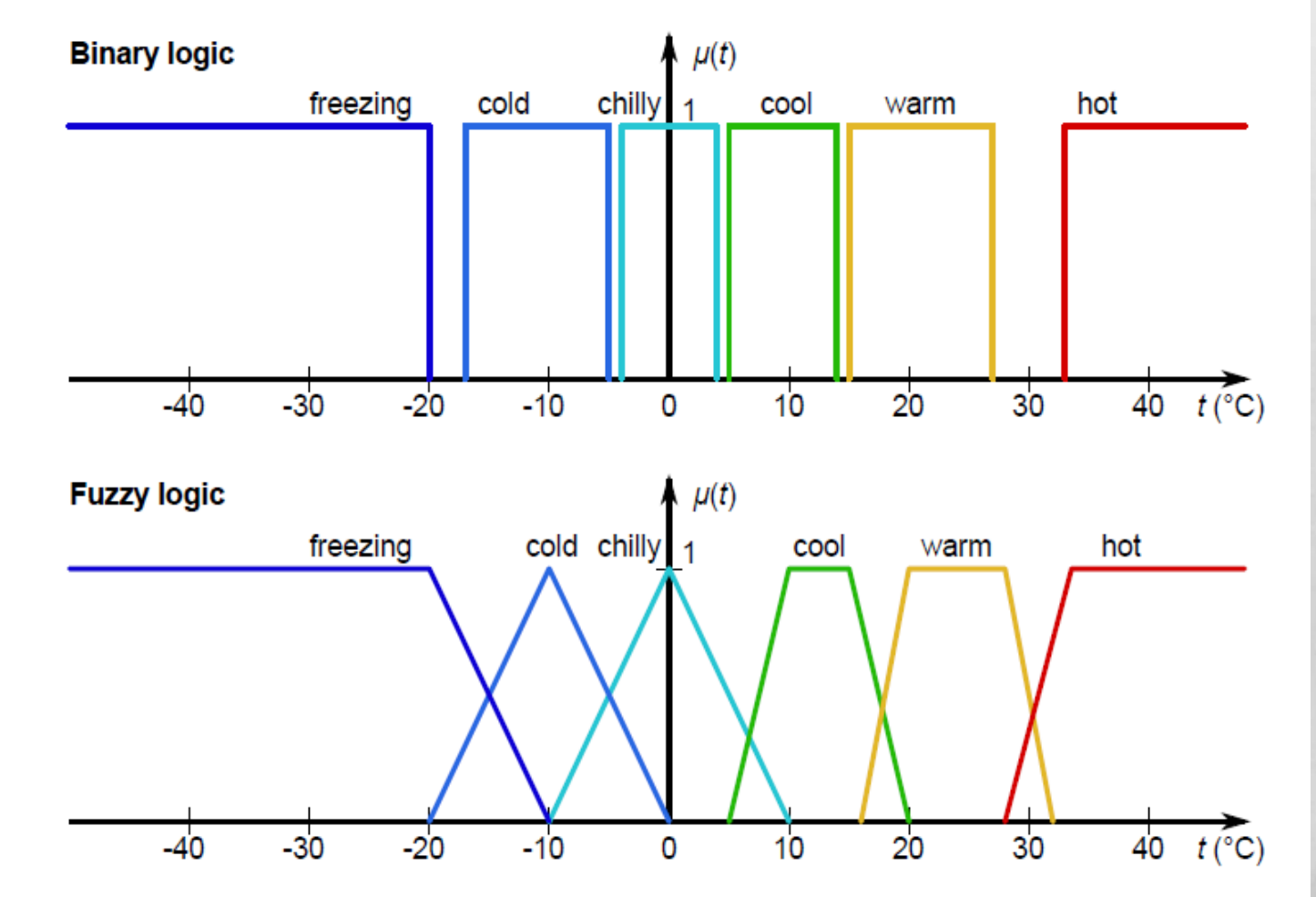
- A = 0.4/Masahiro+0.6/Tsuyoshi+0.8/Takuya+1.0/Saburo+0.9/Masami
- B = 0.3/Masahiro+0.5/Tsuyoshi+0.9/Takuya+0.6/Saburo+0.9/Masami

The OR and AND of A and B are as follows:

- A∪B=0.4/Masahiro+0.6/Tsuyoshi+0.9/Takuya+1.0/Saburo+0.9/Masami
- A∩B =0.3/Masahiro+0.5/Tsuyoshi+0.8/Takuya+0.6/Saburo+0.9/Masami

The Fuzzy Way of Thinking

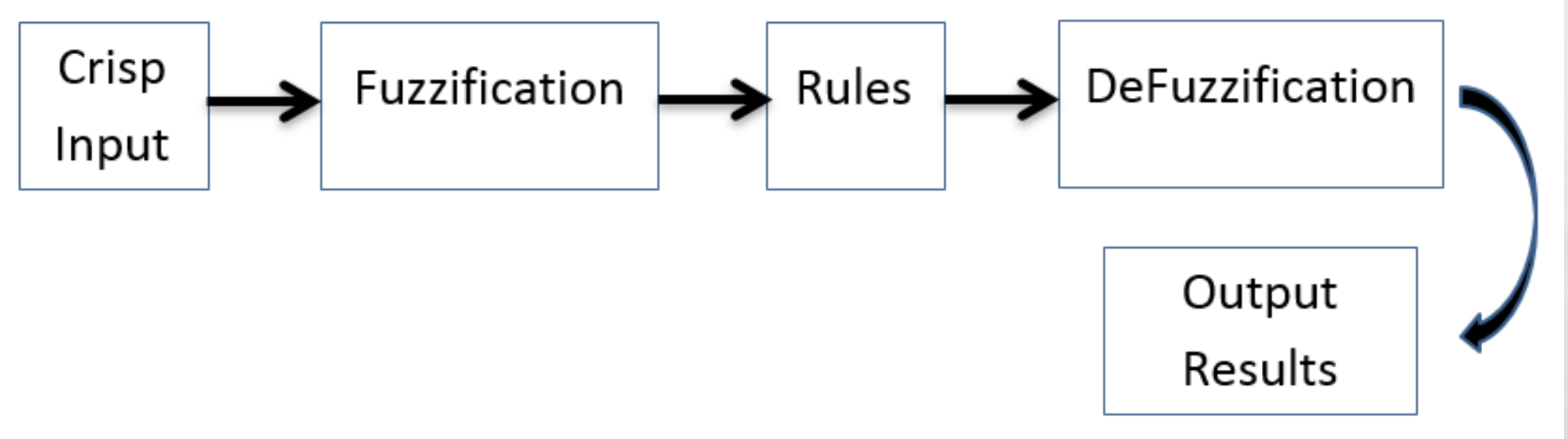
- Binary logic: as we see there is a distance between **freezing** and **cold** this distance we can not decided if it is belonging to freezing or cold.
- Fuzzy logic



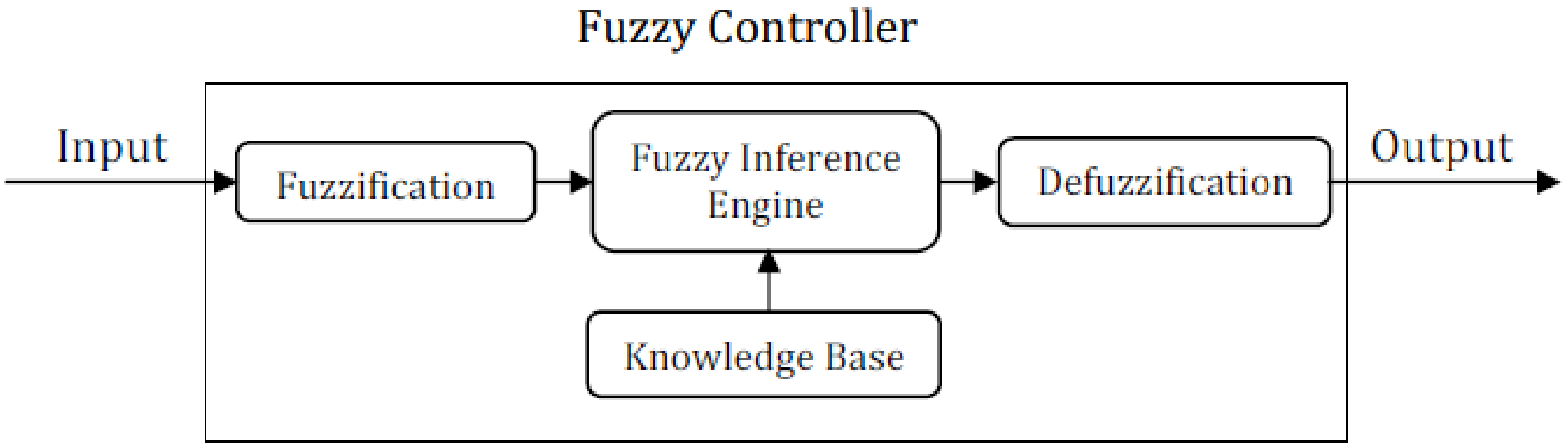
The Concept of Fuzzy Logic

- Fuzzy logic variables have a range of truthfulness from 0 to 1
- Fuzzy logic operates with **linguistic variables**, like “temperature” instead of t(C)
- Each variable has a specific number of **linguistic values**, like “hot” or “cold”
- Fuzzy inference is performed using **linguistic rules**, e.g.
  - IF temperature is cold THEN dress warm
- The linguistic values and their truth degree are quantified using **membership functions** (MF)

Fuzzy System basic components



Fuzzy inference system



Fuzzy Logic Systems Architecture

It has four main parts as shown −

1-Fuzzification Module − It transforms the system inputs, which are crisp numbers, into fuzzy sets. It splits the input signal into five steps such as :-

LP	x is Large Positive
MP	x is Medium Positive
S	x is Small
MN	x is Medium Negative
LN	x is Large Negative

Fuzzy Logic Systems Architecture

2.Knowledge Base − It stores IF-THEN rules provided by experts.

3.Inference Engine − It simulates the human reasoning process by making fuzzy inference on the inputs and IF-THEN rules.

4.Defuzzification Module − It transforms the fuzzy set obtained by the inference engine into a crisp value

The Fuzzy Logic Controller (FLC)

The FLC is a decision-making process and is performed in 4 steps as follows.

1. **Fuzzification** of the input variables transforms the system inputs which are crisp values into fuzzy sets.
2. **Rule evaluation**, taking the fuzzified inputs and evaluate them to the antecedents of the fuzzy rules.
3. **Aggregate conclusions**, the process of unification of the outputs of all rules.
4. **Defuzzification** transforms the fuzzy set obtained by the inference engine into a single crisp value.

The Membership Functions

- Membership functions allow you to quantify linguistic term and represent a fuzzy set graphically. A membership function for a fuzzy set A on the universe of discourse X is defined as $\mu A:X \rightarrow [0,1]$.

- Here, each element of X is mapped to a value between 0 and 1. It is called membership value or degree of membership. It quantifies the degree of membership of the element in X to the fuzzy set A.

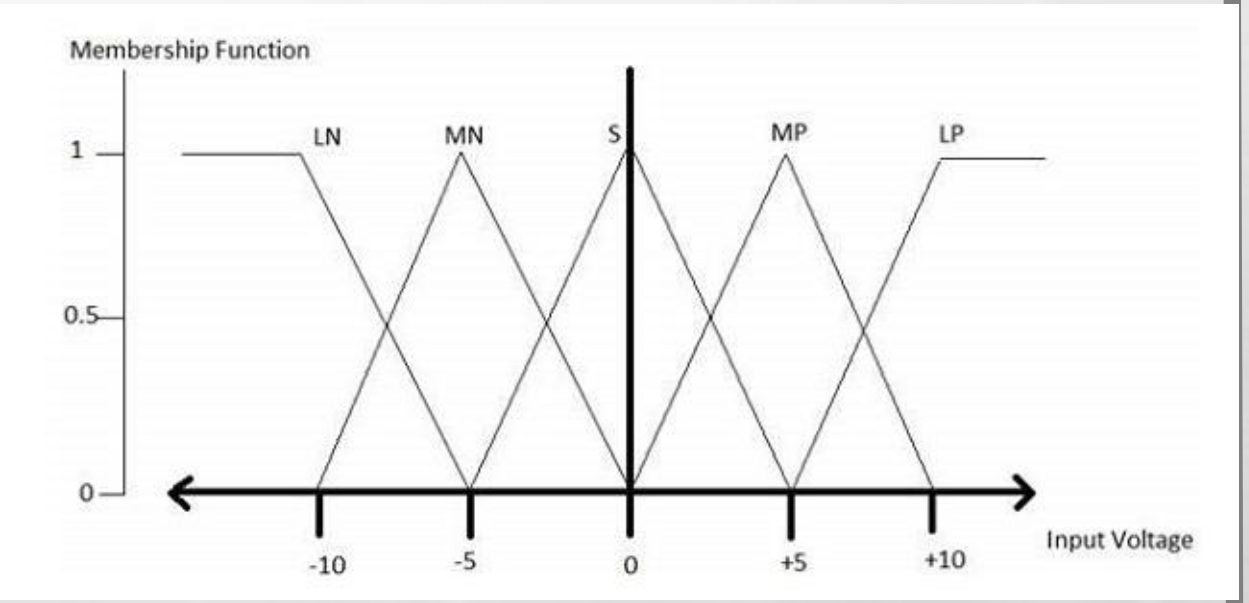
  x **axis represents the universe of discourse.**

  y **axis represents the degrees of membership in the [0, 1] interval**.

- There can be multiple membership functions applicable to fuzzify a numerical value. Simple membership functions are used as use of complex functions does not add more precision in the output.

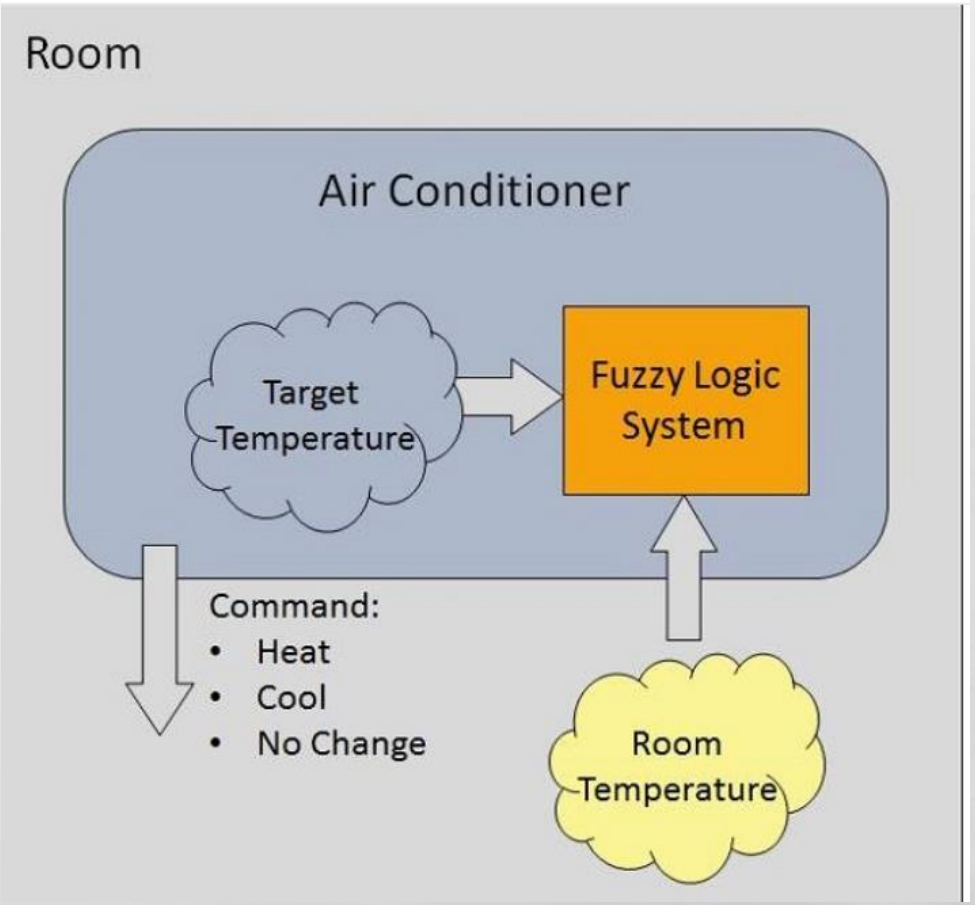
The Membership Functions

- All membership functions for LP, MP, S, MN, and LN are shown as below −
- The triangular membership function shapes are most common among various other membership function shapes such as trapezoidal, singleton, and gaussian.
- Here, the input to 5-level fuzzifier varies from -10 volts to +10 volts. Hence the corresponding output also changes.



Development of Fuzzy logic (EX.1)

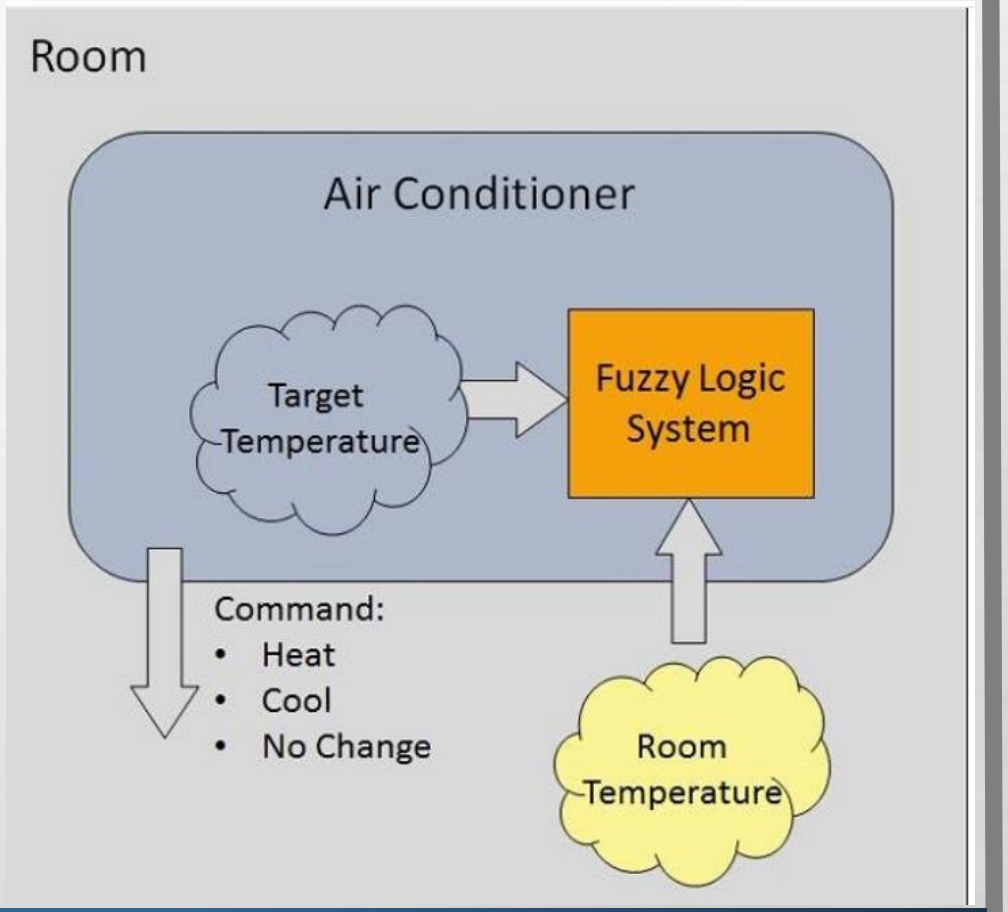
Let us consider an air conditioning system with 5-level fuzzy logic system. This system adjusts the temperature of air conditioner by comparing the room temperature and the target temperature value.



Development of Fuzzy logic (EX.1)

Algorithm

- Define linguistic Variables and terms (start)
- Construct membership functions for them. (start)
- Construct knowledge base of rules (start)
- Convert crisp data into fuzzy data sets using membership functions. (fuzzification)
- Evaluate rules in the rule base. (Inference Engine)
- Combine results from each rule. (Inference Engine)
- Convert output data into non-fuzzy values. (defuzzification)



Development of Fuzzy logic (EX.1)

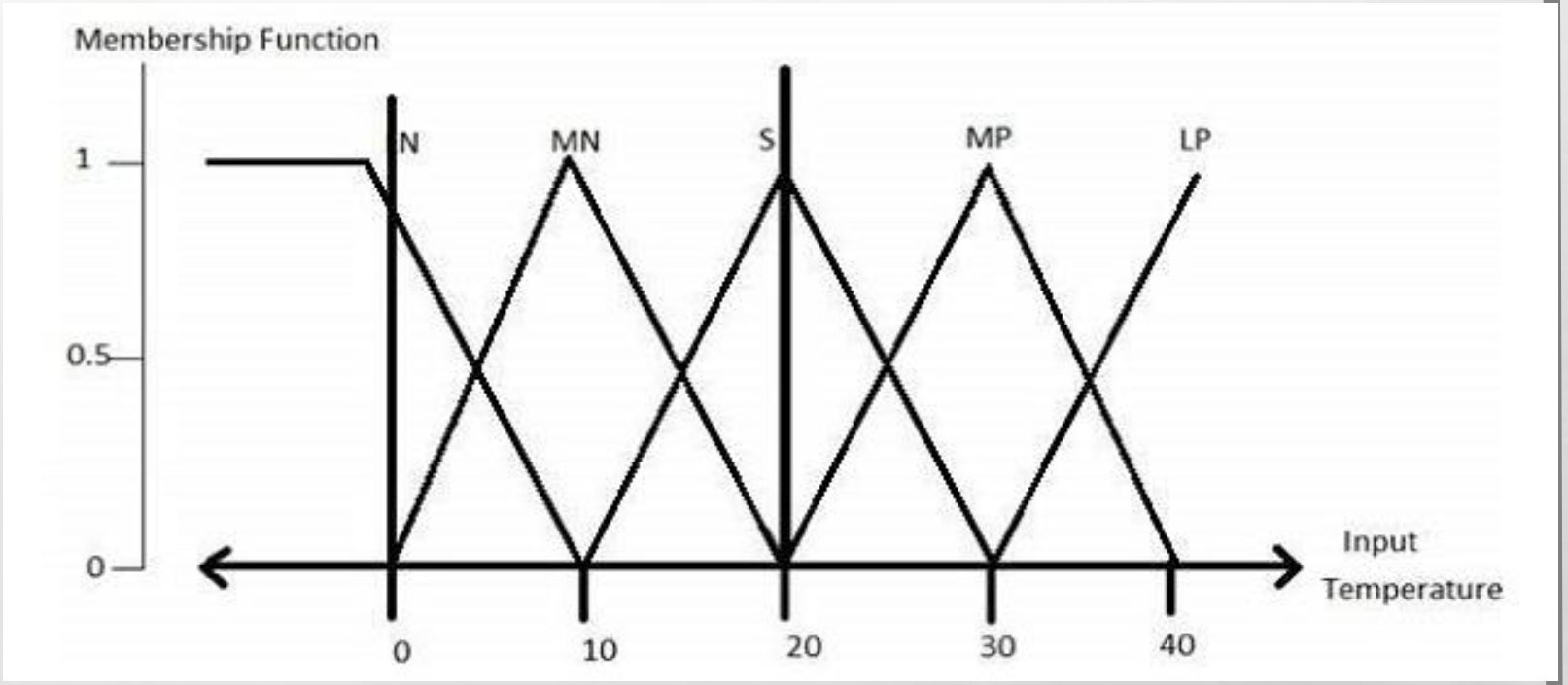
Step 1 − Define linguistic variables and terms

- Linguistic variables are input and output variables in the form of simple words or sentences. For room temperature, cold, warm, hot, etc., are linguistic terms.

Development of Fuzzy logic (EX.1)

Step 2 – Construct membership functions for them

- The membership functions of temperature variable are as shown –



Development of Fuzzy Logic (EX.1)

- Step3 – construct knowledge base rules
- Build a set of rules into the knowledge base in the form of IF-THEN-ELSE structures.

Build a set of rules into the knowledge base in the form of IF-THEN-ELSE structures.

Sr. No.	Condition	Action
1	IF temperature=(Cold OR Very_Cold) AND target=Warm THEN	Heat
2	IF temperature=(Hot OR Very_Hot) AND target=Warm THEN	Cool
3	IF (temperature=Warm) AND (target=Warm) THEN	No_Change

Room Temp. /Target	Very_Cold	Cold	Warm	Hot	Very_Hot
Very_Cold	No_Change	Heat	Heat	Heat	Heat
Cold	Cool	No_Change	Heat	Heat	Heat
Warm	Cool	Cool	No_Change	Heat	Heat
Hot	Cool	Cool	Cool	No_Change	Heat
Very_Hot	Cool	Cool	Cool	Cool	No_Change

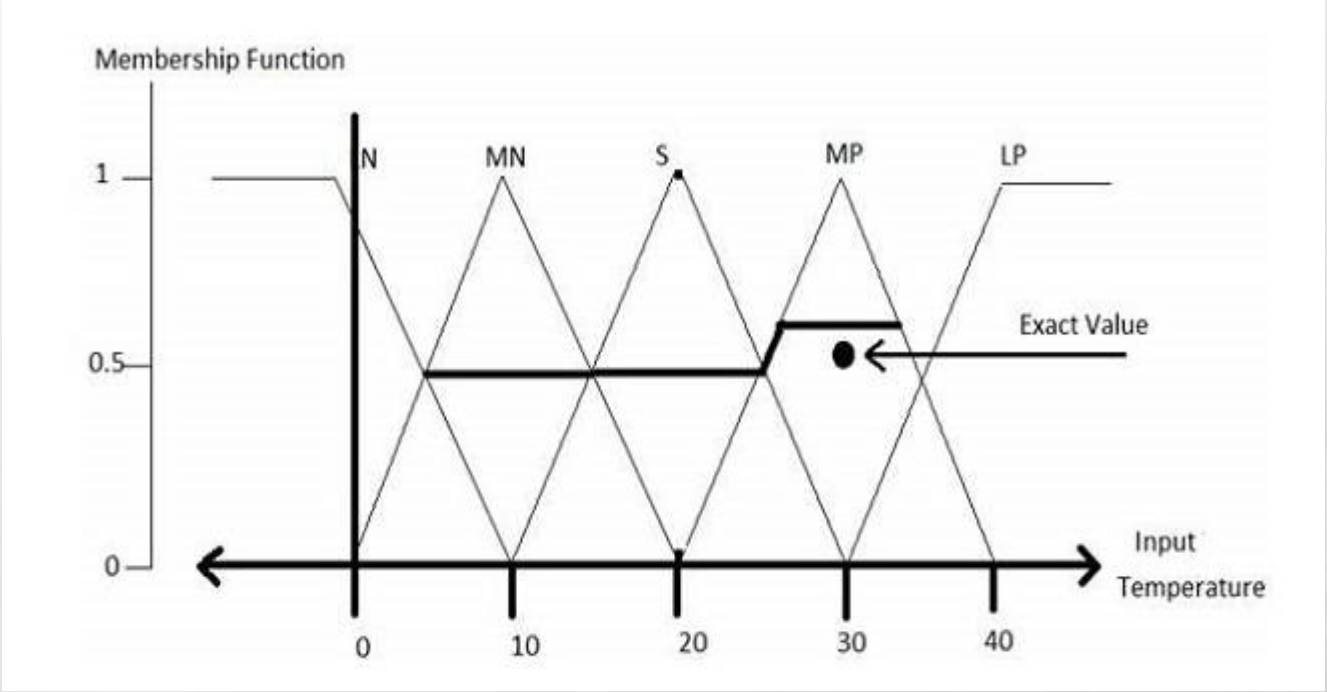
Development of Fuzzy logic (EX.1)

- Step 4 – Obtain fuzzy value
- Fuzzy set operations perform evaluation of rules. The operations used for **OR and AND are Max and Min respectively**. Combine all results of evaluation to form a final result. This result is a fuzzy value.

Development of Fuzzy logic (EX.1)

- Step 5 – Perform defuzzification

Defuzzification is then performed according to membership function for output variable.



Example 2: evaluate a health

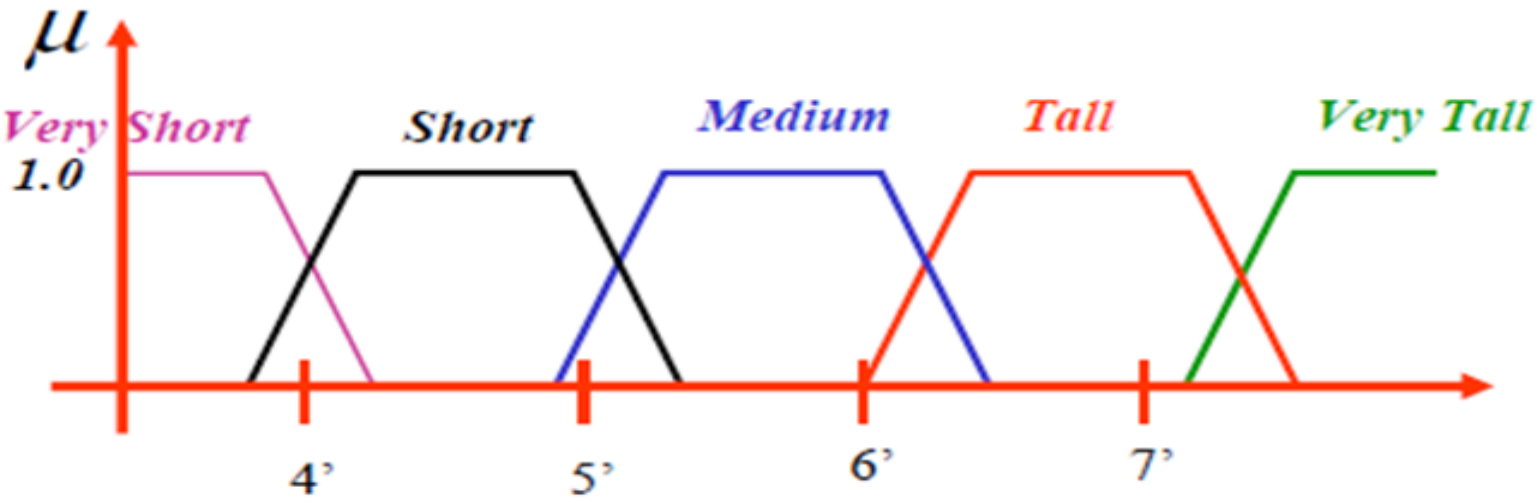
Assume we want to evaluate a **health** of a person based on his **height** and **weight.**

- The input variables are the crisp numbers of a person's height, weight.
- Fuzzification is a process by which a numbers are changes into linguistic words.

Example 2: evaluate a health

Fuzzification of height

- very short,
- short,
- medium,
- tall,
- very tall.

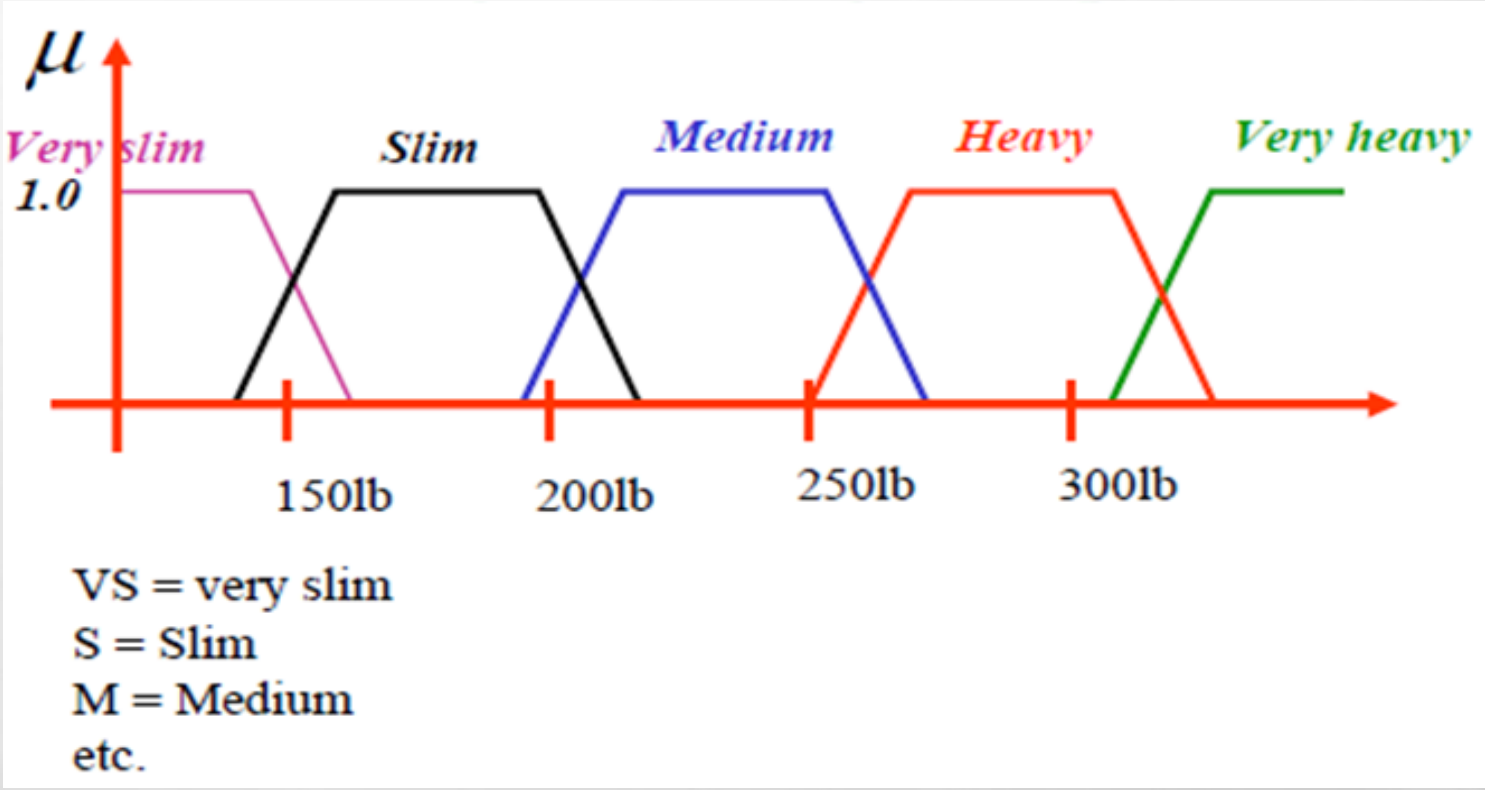


VS = very short
S = Short
M = Medium
etc.

Example 2: evaluate a health

Fuzzification of weight

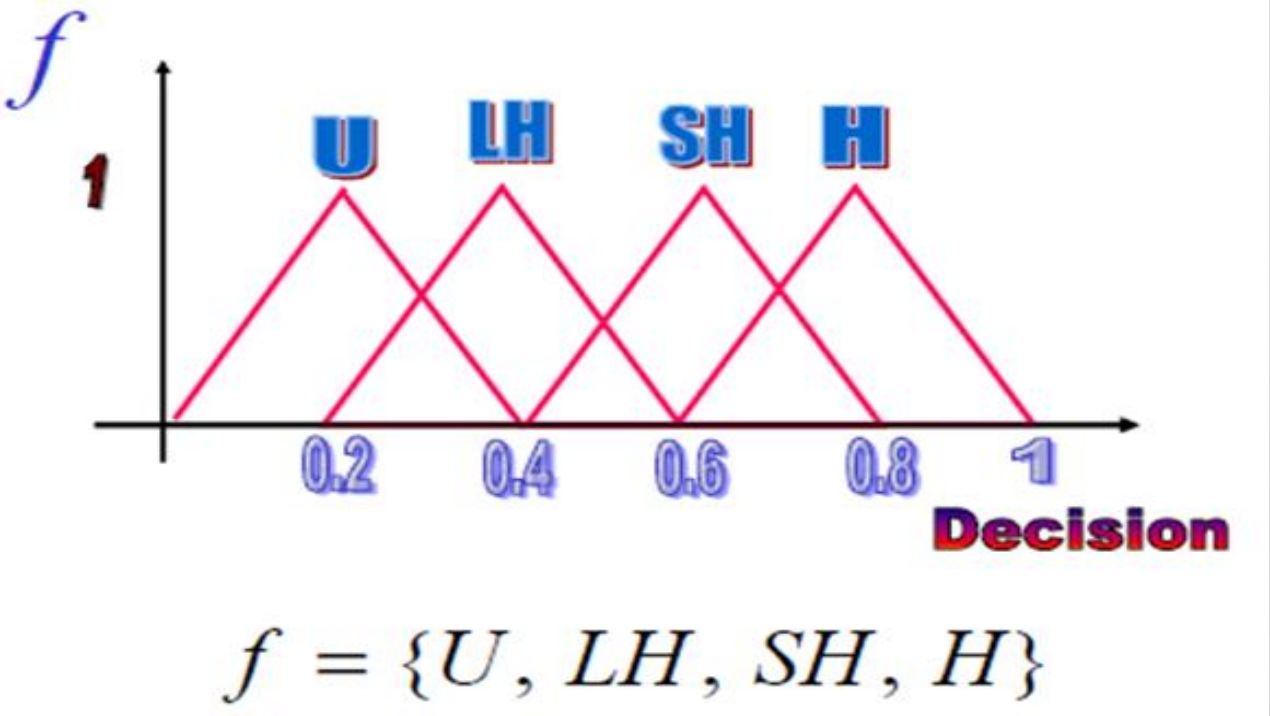
- very slim,
- slim,
- medium,
- heavy,
- very heavy.



Example 2: evaluate a health

Fuzzified Decision

- Healthy(H)
- Somewhat Healthy(SH)
- Less Healthy(LH)
- Unhealthy(UH))



$$f = \{U, LH, SH, H\}$$

Example 2: evaluate a health

Fuzzy Rules Table

Height	Weight: Very Slim	Slim	Medium	Heavy	Very Heavy
Very Short	H	SH	LH	U	U
Short	SH	H	SH	LH	U
Medium	LH	H	H	LH	U
Tall	U	SH	H	SH	U
Very Tall	U	LH	H	SH	LH

Example 2: evaluate a health

For a given person, compute the membership of his/her weight and height.

Example:

- Assume that a person height is 6.1
- Assume that the person weight is 140 LB

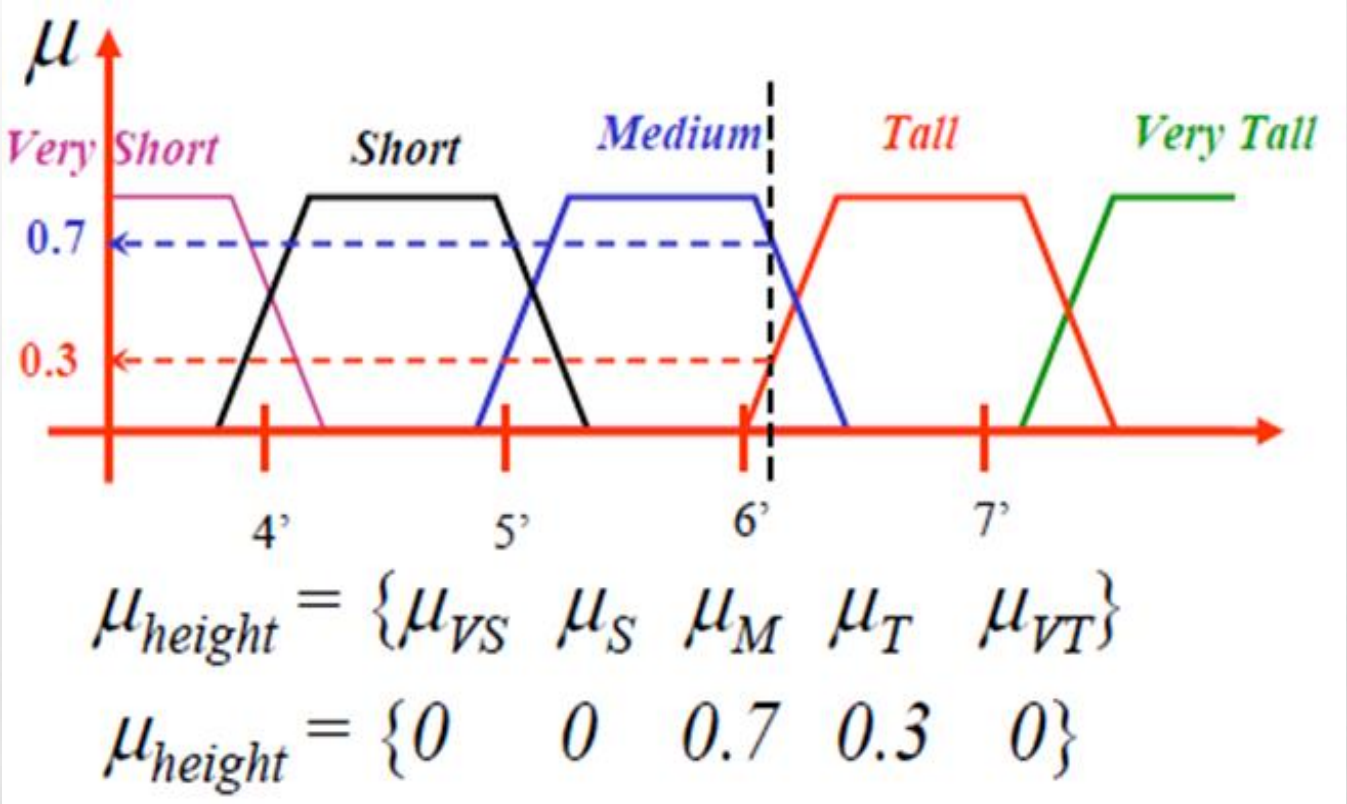
The question is to classify this person to Healthy, Somewhat Healthy,

Less Healthy, or Unhealthy using fuzzy logic.

Solution: evaluate a health

1. The first step is to construct the Membership of height

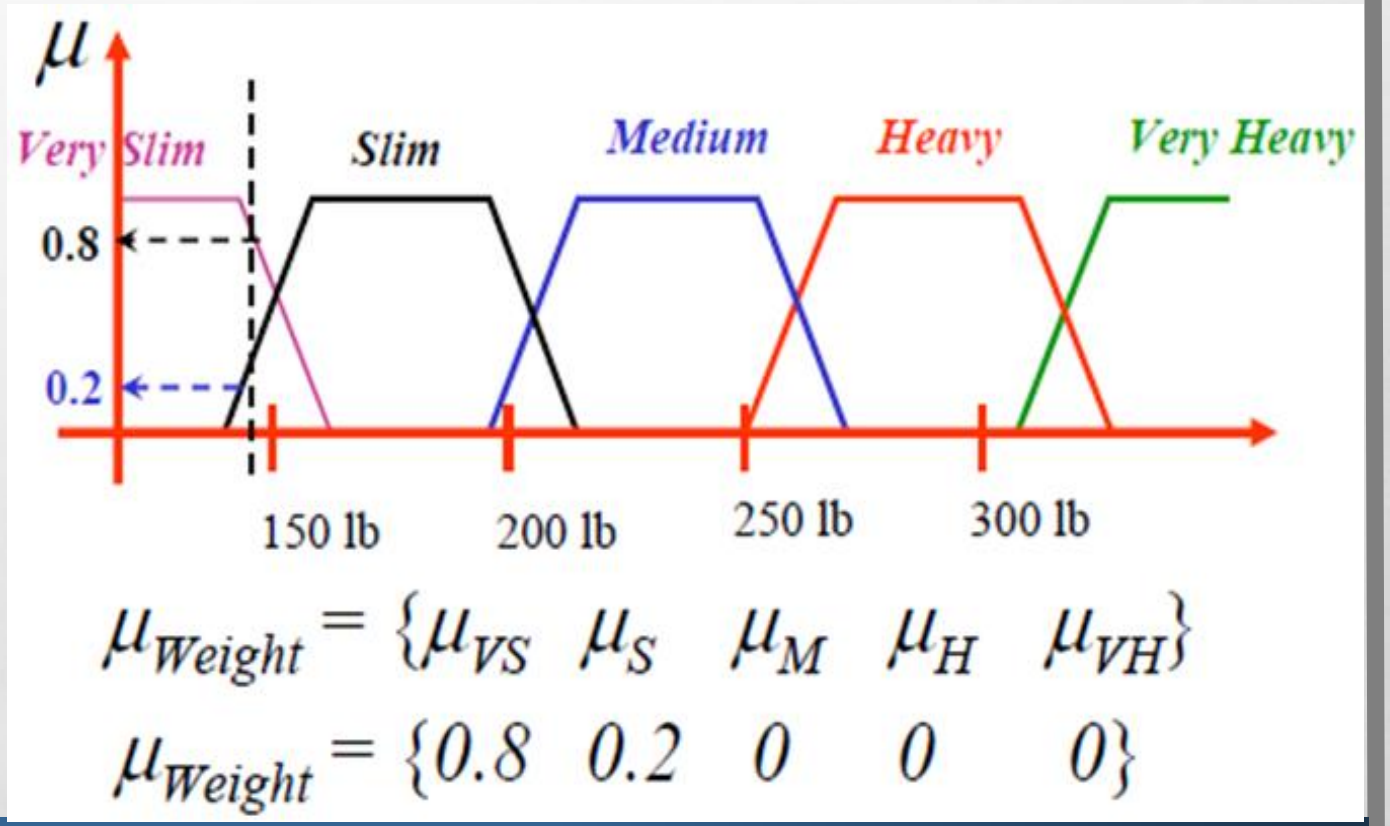
The person height is 6.1



Solution: evaluate a health

2. The first step is to construct the Membership of weight

The person weight is 140



$$\mu_{Weight} = \{\mu_{VS} \quad \mu_S \quad \mu_M \quad \mu_H \quad \mu_{VH}\}$$

$$\mu_{Weight} = \{0.8 \quad 0.2 \quad 0 \quad 0 \quad 0\}$$

Solution: evaluate a health

3. Active Rules

Height \ Weight	Very Slim	Slim	Medium	Heavy	Very Heavy
Very Short	H	SH	LH	U	U
Short	SH	H	SH	LH	U
Medium			H	LH	U
Tall			H	SH	U
Very Tall	U	LH	H	SH	LH

4. Substitute membership values

Height \ Weight	0.8	0.2	Medium	Heavy	Very Heavy
Very Short	H	SH	LH	U	U
Short	SH	H	SH	LH	U
0.7	LH	H	H	LH	U
0.3	U	SH	H	SH	U
Very Tall	U	LH	H	SH	LH

Solution: evaluate a health

5. Perform minimum operation

Height	Weight					
		0.8	0.2	Medium (0)	Heavy (0)	V.Heavy (0)
	V. Short (0)	0	0	0	0	0
	Short (0)	0	0	0	0	0
	0.7	*0.7*	*0.2*	0	0	0
	0.3	*0.3*	*0.2*	0	0	0
	V. Tall (0)	0	0	0	0	0

Solution: evaluate a health

6. Compute decision function.

	Weight	
	0.8	0.2
V. Short (0)	0	0
Short (0)	0	0
0.7	*0.7*	*0.2*
0.3	*0.3*	*0.2*
V. Tall (0)	0	0

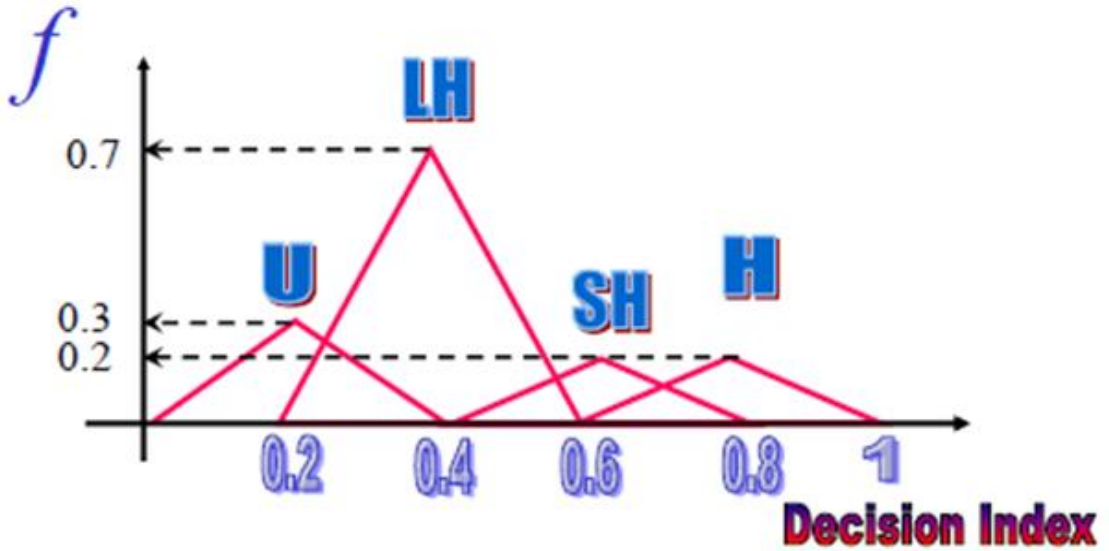
	Weight		
		0.8	0.2
	Very Short	H	SH
	Short	SH	H
	0.7	LH	H
	0.3	U	SH
	Very Tall	U	LH

$$f = \{U, LH, SH, H\}$$

$$f = \{0.3, 0.7, 0.2, 0.2\}$$

Solution: evaluate a health

7. Scaled Fuzzified Decision.



$$f = \{U, LH, SH, H\}$$

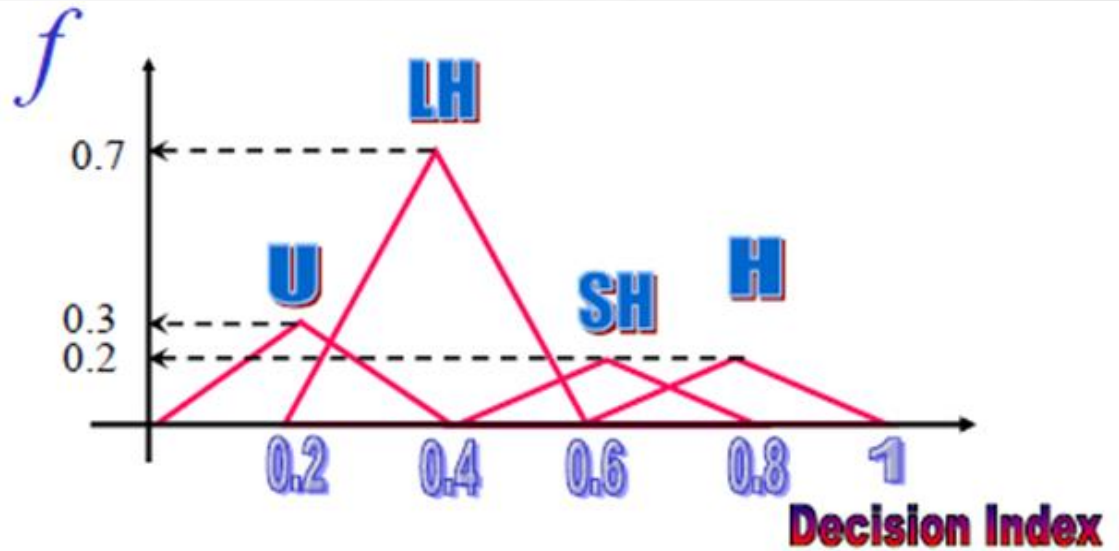
$$f = \{0.3, 0.7, 0.2, 0.2\}$$

Solution: evaluate a health

8. Defuzzification.

a) Maximum method

b) Centroid method



$$f = \{U, LH, SH, H\}$$

$$f = \{0.3, 0.7, 0.2, 0.2\}$$

Solution: evaluate a health

8. Defuzzification->Maximum.

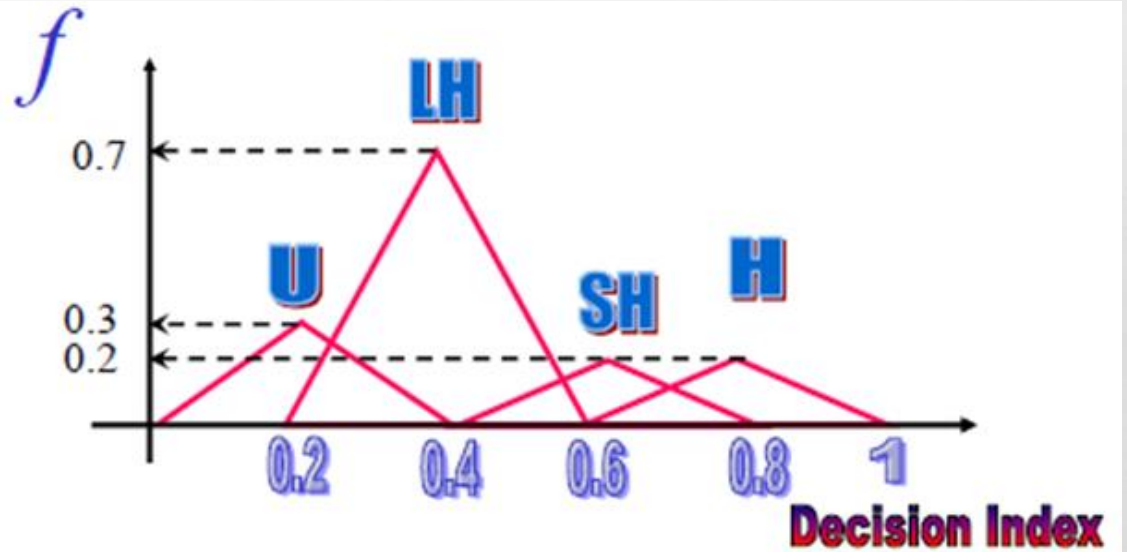
Fuzzy set with the largest membership value is selected.

Fuzzy decision

F = { UH, LH, SH, H}

F = { 0.3, 0.7, 0.2, 0.2}

Final decision (FD)= less Healthy.



$f = \{U, LH, SH, H\}$

$f = \{0.3, 0.7, 0.2, 0.2\}$

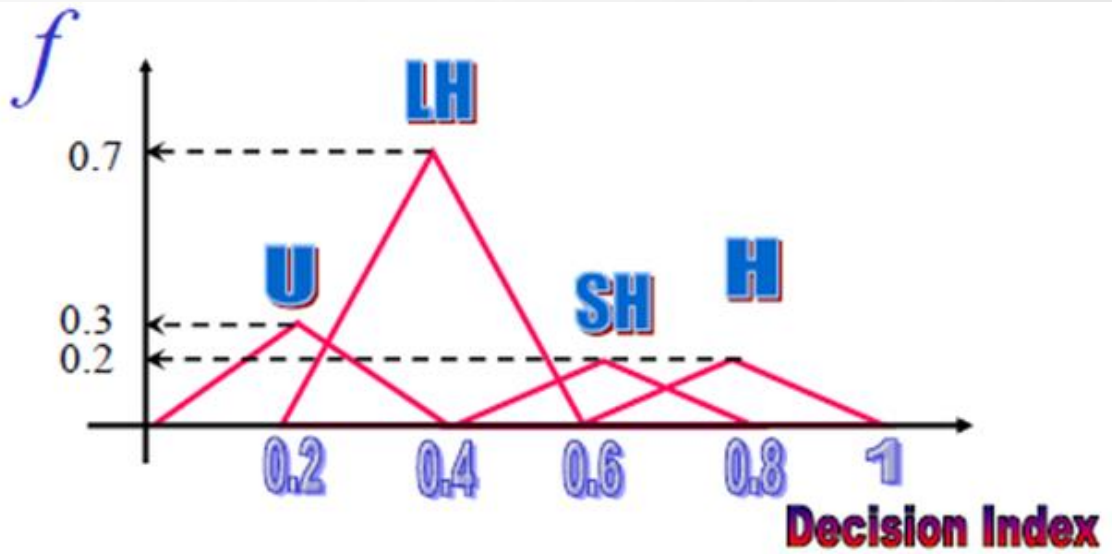
Solution: evaluate a health

8. Defuzzification-> **Centroid**.

$$FD = \frac{\sum \mu D}{\sum \mu} = \frac{\mu_U D_U + \mu_{LH} D_{LH} + \dots \dots}{\mu_U + \mu_{LH} + \dots \dots}$$

$$FD = \frac{0.3 \times 0.2 + 0.7 \times 0.4 + 0.2 \times 0.6 + 0.2 \times 0.8}{0.3 + 0.7 + 0.2 + 0.2} = 0.4429$$

Crisp decision index (D)= 0.4429



$$f = \{U, LH, SH, H\}$$

$$f = \{0.3, 0.7, 0.2, 0.2\}$$

Solution: evaluate a health

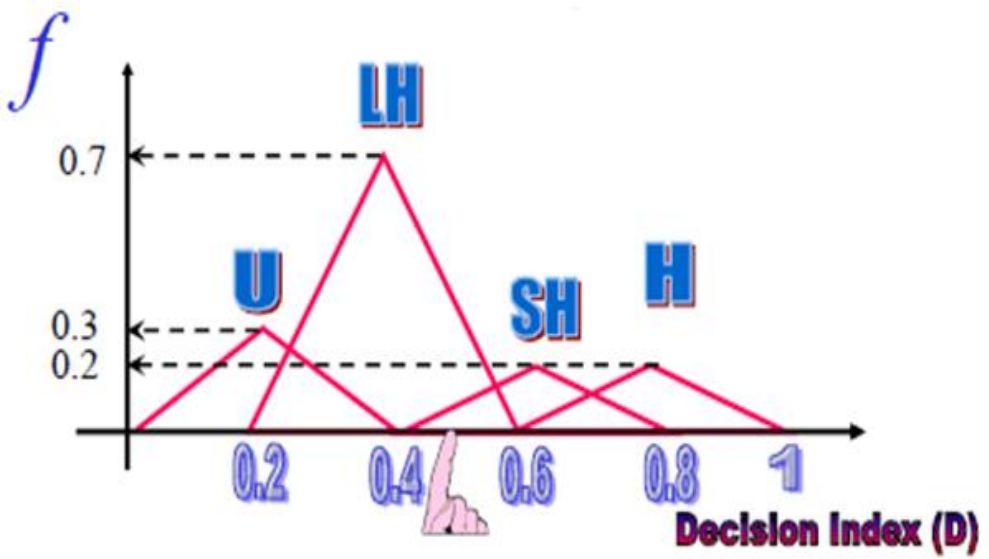
8. Defuzzification-> **Centroid**.

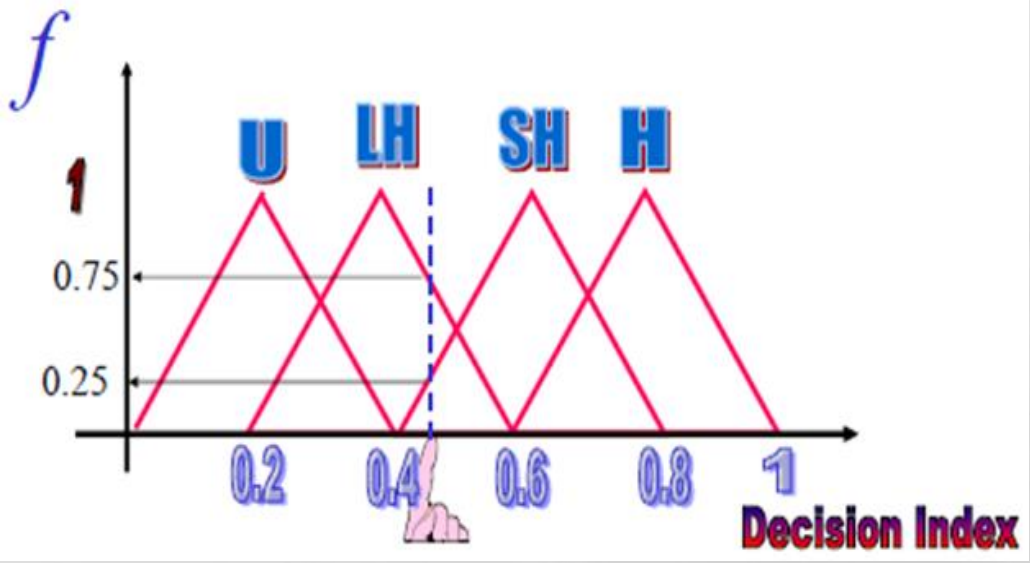
Crisp decision index (D)= 0.4429

Fuzzy decision index (D)

75% in less healthy group

25% in somewhat healthy group.





Disadvantages of FLSs

1. There is no systematic approach to fuzzy system designing.
2. They are understandable only when simple.
3. They are suitable for the problems which do not need high accuracy.

Advantages of FLSs

1. Mathematical concepts within fuzzy reasoning are very simple.
2. You can modify a FLS by just adding or deleting rules due to flexibility of fuzzy logic.
3. Fuzzy logic Systems can take imprecise, distorted, noisy input information.
4. FLSs are easy to construct and understand.
5. Fuzzy logic is a solution to complex problems in all fields of life, including medicine, as it resembles human reasoning and decision making.

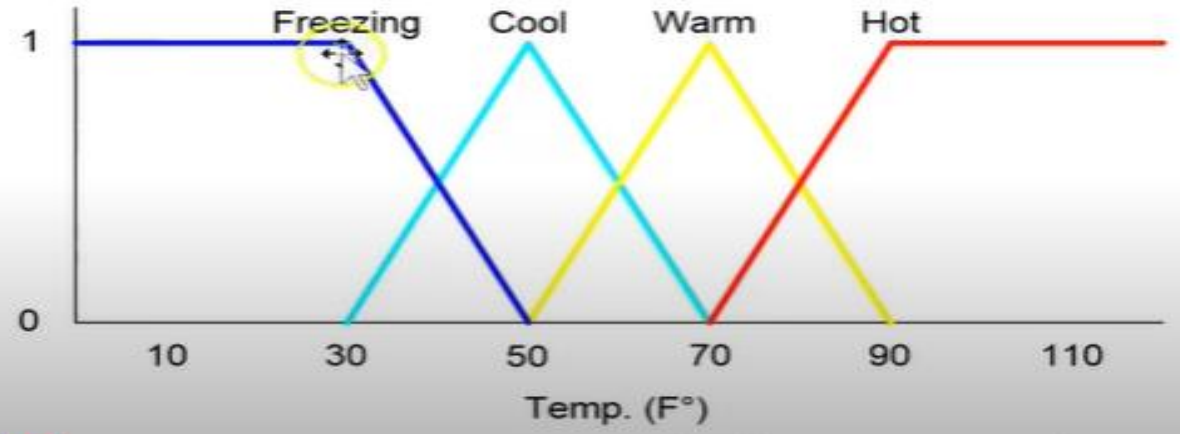
Review Questions

1. Explain the benefit of using the Gaussian theory?.
2. Why Laplace estimator is used?
3. What is the meaning of fuzzy logic, mention 5 different applications of it.
4. Differentiate between Conventional set vs. fuzzy set?
5. What is the meaning of membership function?
6. Explain how fuzzy systems work, and discuss each step.

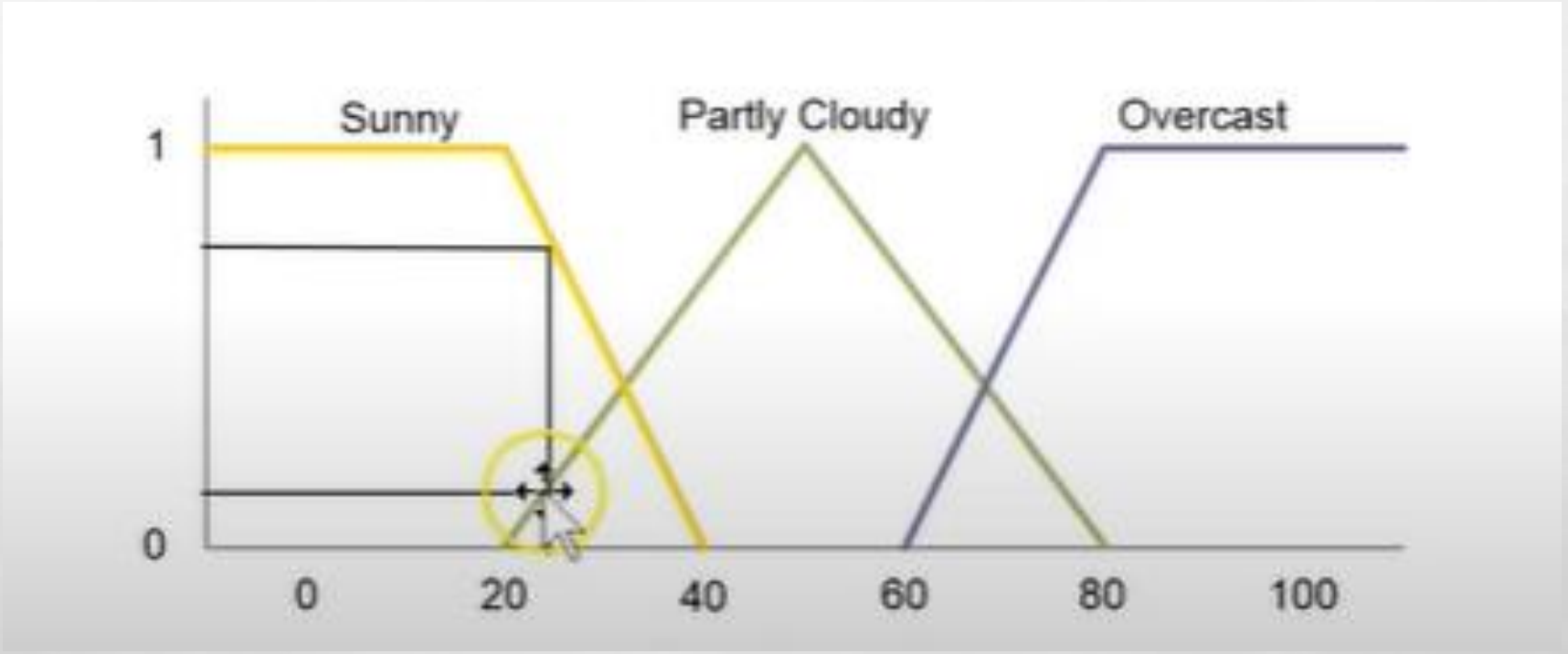
Review Questions

Membership Functions

- Temp: {Freezing, Cool, Warm, Hot}
- Degree of Truth or "Membership"



Review Questions



Review Questions

Rules

- If it's Sunny and Warm, drive Fast

  $\text{Sunny(Cover)} \wedge \text{Warm(Temp)} \Rightarrow \text{Fast(Speed)}$

- If it's Cloudy and Cool, drive Slow

  $\text{Cloudy(Cover)} \wedge \text{Cool(Temp)} \Rightarrow \text{Slow(Speed)}$

- Driving Speed is the combination of output of these rules...

Review Questions

Example Speed Calculation

- How fast will I go if it is
  - 65 F° (=18 °C)
  - 25 % Cloud Cover ?

Review Questions

Use the fuzzy logic (centroid) method to classify a student into Bad (B), Good (G), Very Good (VG) or Excellent (E). The classification will be based on two feature **hours of study** and **focus on studying**. The linguistic variable for both features as the following (Very Low VL, Low L, Medium M, High H, Very High VH).

Decision: B (0, 0.2, 0.4), G (0.2, 0.4, 0.6), VG (0.4, 0.6, 0.8) and E (0.6, 0.8, 1).

Hours of study {VL (-∞, 3, 6), L (3, 6, 9), M (6, 9, 12), H(9, 12, 15), VH (12, 15, ∞)}.

Focus on studying {VL (-∞, 10, 20), L (10, 20, 30), M (20, 30, 40), H(30, 40, 50), VH (40,50, ∞)}.

Classify student has **7** and **24** hours of study and focus on studying respectively. Use the following decision rules table.

	Hours of Study					
Focus on Studying		VL	L	M	H	VH
	VL	B	B	G	G	VG
	L	B	B	G	VG	VG
	M	B	G	VG	E	E
	H	G	G	V G	E	E
	VH	G	VG	E	E	E

Decision Rules Table

Questions

Thank You